

1 Solution — GIAM 3.4

Problem 4

1.1 Problem

It has been conjectured that whenever p is prime, $2^p - 1$ is also prime. Find a minimal counterexample.

1.2 Answer

$p = 11 : \quad 2^{11} - 1 = 2047 = 23 \times 89.$
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1.3 Why It Is Minimal

A *minimal* counterexample requires showing not only that $p = 11$ fails, but that no smaller prime does. There are four smaller primes, and all four succeed:

p	$2^p - 1$	
2	3	prime
3	7	prime
5	31	prime
7	127	prime
11	2047	composite = 23×89

Since $23 \times 89 = 2047$ exactly and both factors are prime, $p = 11$ is a genuine counterexample, and it is the least one.

1.4 The Picture

The table below runs through every prime exponent up to **41**, and the bands around it record the true theorem the conjecture inverts and the congruence that makes the factor of **2047** easy to find.

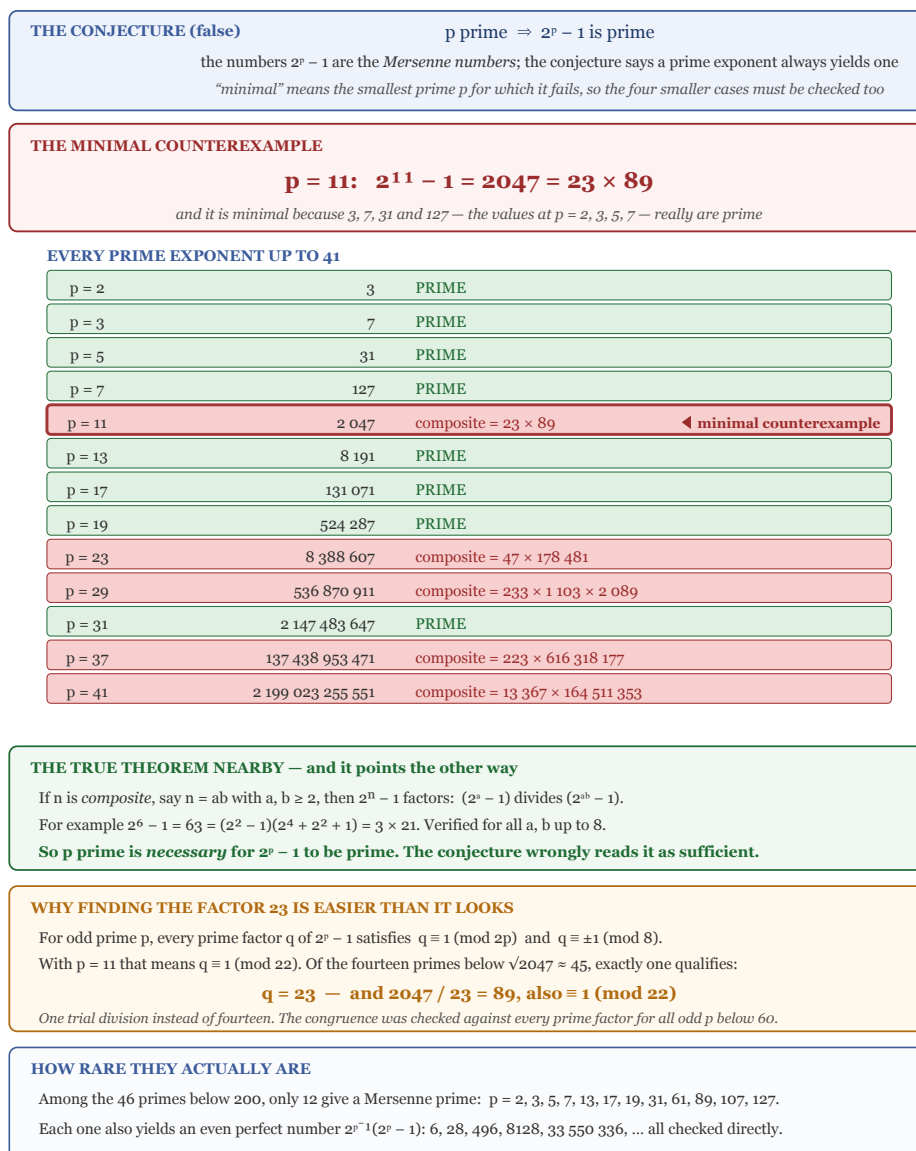


Figure 1.1: A diagram in five parts. The first band states the false conjecture: p prime implies 2 to the p minus 1 is prime. The numbers 2 to the p minus 1 are the Mersenne numbers, and the conjecture says a prime exponent always yields one. Minimal means the smallest prime p for which it fails, so the four smaller cases must be checked too. The second band gives the minimal counterexample: p equals 11 , and 2 to the eleventh minus 1 equals 2047 equals 23 times 89 . It is minimal because $3, 7, 31$ and 127 — the values at p equals $2, 3, 5$ and 7 — really are prime. The centre is a table of every prime exponent up to 41 . Green prime rows: p equals 2 giving 3 , p equals 3 giving 7 , p equals 5 giving 31 , p equals 7 giving 127 , p equals 13 giving 8191 , p equals 17 giving 131071 , p equals 19 giving 524287 , and p equals 31 giving 2147483647 . Red composite rows: p equals 11 giving 2047 equals 23 times 89 , outlined heavily and marked as the minimal counterexample; p equals 23 giving 8388607 equals 47 times 178481 ; p equals 29 giving 536870911 equals 233 times 1103 times 2089 ; p equals 37 giving 137438953471 equals 223 times

616318177; and p equals 41 giving 2199023255551 equals 13367 times 164511353. A band below states the true theorem nearby, which points the other way: if n is composite, say n equals ab with a and b at least 2, then 2 to the n minus 1 factors, because 2 to the a minus 1 divides 2 to the ab minus 1. For example 2 to the sixth minus 1 equals 63 equals the product of 2 squared minus 1 and 2 to the fourth plus 2 squared plus 1, that is 3 times 21; verified for all a and b up to 8. So p prime is necessary for 2 to the p minus 1 to be prime, and the conjecture wrongly reads it as sufficient. The next band explains why finding the factor 23 is easier than it looks: for odd prime p , every prime factor q of 2 to the p minus 1 satisfies q congruent to 1 mod $2p$ and q congruent to plus or minus 1 mod 8. With p equal to 11 that means q congruent to 1 mod 22, and of the fourteen primes below the square root of 2047, about 45, exactly one qualifies: q equals 23, and 2047 divided by 23 is 89, also congruent to 1 mod 22. One trial division instead of fourteen; the congruence was checked against every prime factor for all odd p below 60. The final band shows how rare Mersenne primes are: among the 46 primes below 200, only 12 give one, at p equal to 2, 3, 5, 7, 13, 17, 19, 31, 61, 89, 107 and 127. Each one also yields an even perfect number 2 to the p minus 1 times the quantity 2 to the p minus 1: 6, 28, 496, 8128, 33550336 and so on, all checked directly.

1.5 Remark — The Conjecture Inverts a Real Theorem

There *is* a theorem connecting the primality of n to that of $2^n - 1$, and the conjecture is that theorem read backwards.

Theorem. *If n is composite, then $2^n - 1$ is composite.*

Proof. *Write $n = ab$ with $a, b \geq 2$. The factorisation*

$$x^b - 1 = (x - 1)(x^{b-1} + x^{b-2} + \cdots + x + 1)$$

applied with $x = 2^a$ gives

$$2^{ab} - 1 = (2^a - 1)(2^{a(b-1)} + \cdots + 2^a + 1).$$

Both factors exceed 1 because $a \geq 2$ and $b \geq 2$, so $2^n - 1$ is composite. ■

For instance $2^6 - 1 = 63 = (2^2 - 1)(2^4 + 2^2 + 1) = 3 \times 21$. The divisibility $(2^a - 1) \mid (2^{ab} - 1)$ was verified for all $a, b \leq 8$.

Contrapositively: **if $2^n - 1$ is prime then n is prime**. That is the true statement. The conjecture in this problem asserts the converse, and the converse of a true implication is a separate claim entirely — precisely the confusion this section is designed to expose. Primality of p is *necessary* for $2^p - 1$ to be prime, not *sufficient*.

1.6 Remark — Finding the Factor Without Guessing

Factoring 2047 by hand looks like fourteen trial divisions, one for each prime below $\sqrt{2047} \approx 45$. It is actually one, because the factors of Mersenne numbers are heavily constrained.

For an odd prime p , every prime factor q of $2^p - 1$ satisfies

$$q \equiv 1 \pmod{2p} \quad \text{and} \quad q \equiv \pm 1 \pmod{8}.$$

The first congruence comes from Fermat: $2^p \equiv 1 \pmod{q}$ means the order of 2 modulo q divides p , and since the order is not 1 it equals p ; so $p \mid q - 1$, and $q - 1$ is even, giving $2p \mid q - 1$. The second follows from quadratic reciprocity. Both were checked against every prime factor of $2^p - 1$ for all odd primes $p < 60$.

With $p = 11$ this says $q \equiv 1 \pmod{22}$, so the candidates are 23, 67, 89, 199, ... — and below 45 there is exactly one:

$$q = 23, \quad 2047 = 23 \times 89,$$

with the cofactor $89 = 4 \cdot 22 + 1$ obeying the same congruence, as it must. This is why Mersenne numbers were the testing ground for primality records long before computers: the search space for factors is tiny, and there is a dedicated primality test for them besides — the Lucas–Lehmer test, which settles $2^p - 1$ in about p squarings and no trial division at all.

1.7 Remark — How Sparse the Mersenne Primes Are

Among the 46 primes below 200, only 12 produce a Mersenne prime:

$$p = 2, 3, 5, 7, 13, 17, 19, 31, 61, 89, 107, 127.$$

So roughly a quarter of small prime exponents work, and the proportion falls away sharply as p grows. Notice too that the failures are not confined to $p = 11$: the table shows $p = 23, 29, 37, 41$ all failing, with factorisations that grow rapidly in size.

As of writing, only 52 Mersenne primes are known in total, the largest being $2^{136279841} - 1$ — found by the distributed GIMPS project in 2024 and running to over 41 million digits. Whether there are infinitely many is a long-standing open problem, as is whether infinitely many $2^p - 1$ are composite. Both are believed true and neither is proved.

1.8 Remark — Why Anyone Cared: Perfect Numbers

The conjecture was not idle. Mersenne primes are in exact correspondence with the even **perfect numbers** — those equal to the sum of their proper divisors.

Euclid–Euler Theorem. *An even number N is perfect if and only if $N = 2^{p-1}(2^p - 1)$ with $2^p - 1$ prime.*

The first few, each verified by summing divisors directly:

p	$2^p - 1$	$N = 2^{p-1}(2^p - 1)$
2	3	6
3	7	28
5	31	496
7	127	8128
13	8191	33 550 336

Note the gap: there is no perfect number between 8128 and 33 550 336, and the reason is exactly the counterexample of this problem — $p = 11$ produces no Mersenne prime, so it produces no perfect number either. The conjecture, had it been true, would have supplied a perfect number for every prime. Its failure at 11 is directly visible as the missing

fifth perfect number in the classical list, a gap that puzzled mathematicians from antiquity until Hudalricus Regius found the factorisation $2047 = 23 \times 89$ in 1536.

1.9 Remark — Mersenne’s Own List Was Wrong Too

Marin Mersenne asserted in 1644 that $2^p - 1$ is prime exactly for

$$p = 2, 3, 5, 7, 13, 17, 19, 31, 67, 127, 257$$

and composite for all other $p \leq 257$. He had already dropped $p = 11$, so he was not making this problem’s error. But his list contains two false entries — $p = 67$ and $p = 257$ give composites — and omits three genuine ones at $p = 61, 89, 107$.

The $p = 67$ case has a famous coda. In 1903 Frank Nelson Cole gave a lecture to the American Mathematical Society consisting entirely of computing $2^{67} - 1$ on one side of a blackboard, then multiplying $193\,707\,721 \times 761\,838\,257\,287$ on the other. The two agreed; he sat down without a word to a standing ovation. It remains the cleanest illustration of what a counterexample is: no argument required, just an instance.